

SYSC 4907: 4th Year Project

Fall Term Info Session

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The 4th Year Project

- An example of good engineering analysis and design that demonstrates your engineering skills
 - Apply lessons learned from courses
- At least 200 hours ... average of 8 hours per week
- Work under the supervision of a faculty member
 - Meetings, milestones, applying engineering processes
- But, the initiative for keeping your project on schedule must come from you
- Good Engineering also requires communication & documentation



- Joy
- Freedom
- Teamwork
- Sky is the limit

The 4th Year Project (cont.)

- It is **NOT** a requirement that you build something that is
 - “innovative”
 - “unique”
- It IS important that you follow a proper engineering approach, **applying, honing, integrating and extending previously acquired knowledge and skills in a major engineering design project.**
 - There needs to be a clear relationship between your degree program and the project
 - You are to use your previously acquired knowledge and skills
 - This is NOT the place to learn a brand-new set of skills
 - The process you follow needs to be documented and will form a major part of your evaluation

Information and Resources

- All information about the project is on the web.
 - www4.sce.carleton.ca/courses/sysc-4907
 - Links : Notices, FAQ, Forms and Deadlines
 - Read the website before asking questions.
 - Note: need access to computer on campus or **Carleton VPN** to access the course website
- Occasional emails will be sent to your *cmail* account, the one you used for registration on the project website.

Getting Started: Registration

- **Please** ensure you are in the correct course.
- SYSC-4907 or ELEC-4907 ??
 - SE and BioMed: SYSC-4907, even if project is external to department.
 - CSE and CE: It matters who your SUPERVISOR is.
 - SYSC-4907: ANY student with a SYSTEMS supervisor
 - ELEC-4907: ANY student with an ELECTRONICS supervisor
 - (PS: You're in the wrong meeting!)
- Must register both fall and winter
 - Oral presentation/poster fair not scheduled and marks not released unless all fees are paid.
- Incorrect registration = delayed graduation

Communications: Group Numbers

- All communications with the coordinator must be done using **your email account**.
 - AND by sending email to ug-projects@sce.carleton.ca, not the project coordinator's individual email account
- Please include your **group identifier** assigned when you joined a project, in all correspondence.

Deadlines (tentative): The Year in Brief

- Fall Term
 - Friday, Oct 28: **Proposal** (submit on-line and to the supervisor)
 - More later on proposal content
 - Friday, Dec 6:
 - **Oral Presentation choice made on-line**
 - **Progress Report** (submit online and to the supervisor)
- Winter Term
 - January 20-24: Oral Presentations
 - March 21: **Final Report Draft** (submit to the supervisor)
 - March 28: **Poster Fair**
 - **Wednesday, April 8:** Final Report (submit on-line and to the supervisor) and Video (**submit on-line.**)
 - Missed deadline means F.

Supervisor versus Coordinator

- Coordinator (non-technical)
 - Looks after overall administration of the projects
 - Organizes departmental-wide events
 - Has no involvement in the project itself (not a co-supervisor)
- Supervisor (technical)
 - Responsible for regular supervision of progress
 - Responsible for informing student of all project requirements
 - Governs the format of the project deliverables, within the guidelines stated for the project.
 - Responsible for providing any resource required by project
 - Responsible for evaluating student and the project deliverables

Project Resources

- Computer accounts are the same SCE-accounts used in your other Systems courses.
 - You may have to re-activate it.
 - Tied to your Connect account.
- Special requests regarding your account must be made by your **supervisor** to Daren Russ
 - Extra memory, group or administrator rights
 - Create a ticket: service@sce.carleton.ca
 - Learn a valuable lesson early: Treat your support staff well!

Project Resources

- You may need to/want to buy specific equipment for your project – who pays?
 - Option 1: Instead of having students purchase equipment, purchasing will be done by our technicians upon request by students and will approval from their supervisor. More specific instructions will be shared early next week. All groups have a \$500 budget.
 - Option 2: your supervisor (from his or her research funding)
 - KEFC funding information from CSES (Carleton Student Engineering Society):
<https://docs.google.com/document/d/1bpGk9-UFOWRjgyGifCEWqzA8HBI49BYRJ7XsOK8QD0w/edit>
(Check the requirements and timeline – open now!)
- The link to submit a purchase request:
<https://sysc.atlassian.net/servicedesk/customer/portal/3>

Project Deliverables

- What is a **Proposal**?
- What is a **Progress Report**?
- What can you do with **Oral Presentation**?
- What is the **Poster Fair**?
- How big should my **Final Report** be?
- For expectations and formats of deliverables :
 1. **General** guidelines and suggestions are posted on the project website called “Project Deliverables”
 2. Your **supervisor** has the final say. S/he is marking it.

Project Grading

- Two facets of Performance:
 - Technical achievements, including documentation
 - Professional conduct
- Different procedure from other courses
 - No course breakdown is given; grade is an amalgam of your year's performance.
 - Grade assigned by whole department
 - You have two representatives
 - Supervisor: Biased “insider”
 - Second Reader: Objective “outsider”; largely technical evaluation
 - Note that other faculty members may be involved if needed
- Governed by published departmental Grade Expectations

Project Grading

- A Summary of the Project's Deliverables
 - Project Proposal
 - Progress Report
 - Oral Presentation
 - Poster Fair
 - Final Report (See below)
 - Project-Specific Deliverables (other items that the supervisor has required)

Faculty Grading Guidelines: Project

Quality of Engineering Project	D	C	B	A
Engineering Techniques Used				
Requirements and design (the application of engineering sciences)				
Implementation				
Testing				
Evaluation and analysis				

Grading Guidelines: Final Report

- Statement of Objectives
- General Organization (title page, sequence of chapters, references, appendices, etc.)
- Technical Writing and Illustrations (correct English, spelling, clarity, conciseness, accurate and complete)
- Adequate Background Information (preparatory information, literature survey)
- Conclusions and recommendations
- Quality of Final Engineering Product
- Engineering Techniques Used (use of modern concepts and methods, systematic engineering methods)
- Engineering Design (quality of design, functional, adaptable)
- Implementation
- Testing/Experimental Data Used (completeness and accuracy)
- Engineering evaluation and analysis (analysis of the problem & deriving requirements, analysis of data presented, critical review of final product)
- Overall: demonstrated skills in pursuing a proper engineering process

Final Grading: Student

Individual Performance Assessment	D	C	B	A
Initiative				
Organization				
Conduct of Work				
Learning Experience				
Teamwork				

Grade Expectations (Posted)

A: Outstanding achievement

B: Good, solid professional engineering work.

B+ Expected “typical” grade for a successful project.

C: Below-average performance

C- minimal acceptable grade average for graduation.

D: Performance is below graduation standards

D- simply a passing mark for the course credit.

F: is given to those must repeat the course.

“I’ve worked long and hard. I deserve an A”

Getting Down to Work

- How much work is expected from me?
 - Rough rule of thumb: **minimum 8 hours per week.**
- How can I **earn** a good mark on the project?
 - Must contain all engineering elements: requirements analysis, design, implementation, testing and documentation.
 - Must reflect the methodologies taught in courses.
 - The best projects are marked by creativity and initiative.
- Bottom Line: The project is your responsibility.
 - Supervisors can only remind/coach you/team and provide critics for improvement.

Getting Down to Work

- How often should/can I meet with my supervisor?
- When should I start meeting with my supervisor?
 - ASAP once the term starts
- What do I do in our meeting?
- I've got 3 midterms, 5 labs. I can't work on my project.

Getting Down to Work

- I can't get hold of my supervisor.
 - Come talk with me.
- My partner is stupid/lazy ...
 - This is part of the team experience.
 - Understand where your team mate is coming from and confirm that the problem is with the partner (and not you...)
 - Try to resolve it within the group first
 - Talk with your supervisor but in a professional manner.
 - Wait until you have calmed down.
 - Prepare concrete examples.
 - Prepare constructive solutions (re-allocate duties, devise a plan for you to continue despite missing your partner's piece)
 - Show leadership by communicating, compromising.

Deliverables: Group Projects

- For all deliverables, there is a choice between individual and group reports
 - Each individual may write their own OR
 - A common group portion is prepared and shared by all members, followed by individual contributions.
- It is the **supervisor's** decision with the following stipulations:
 - **Each individual must make equal contributions to the project in each phase.**
 - **The author of the portions of the final report must be clearly described.**

Fall Deliverable: Project Proposal

- **Due: Oct 28**
- Purpose:
 - To convey the objectives and the engineering challenges of the proposed project
 - To present a well-developed and justified plan for the work overall and to be done by each member
- **New since 2020/21:**
 - Deadline moved back (though supervisors may require an earlier draft at the end of September)
 - **Requires significant effort**
 - Can become part of the final report

Required Proposal Components

- The title of the project, your names and student numbers, and your supervisor's name.
- A clear statement of the **objectives** of the project.
 - Measurable functional and non-functional requirements
 - How to measure progress towards these objectives.
- A **brief background** of the project: what has been done to address the problem, what the state of the art is, etc.
- A brief description of what **your plan** is.
- A discussion of how the project relates to the degree program of each student.
- How does the group, collectively, have the skills required to undertake the project.
- A description of the **method(s)** you are going to use in solving the problem and how they relate to knowledge you acquired in your degree program to-date.
- A proposed **timetable** for completion of the project
- A discussion of **possible project risks** and mitigation strategies.
- A list of special components and facilities that you require.
- A list of **references**, also proper **citations** in the text

Proposal: Sample Milestones

1. Load and run last year's code into FPGA
2. Finish design for O/S changes.
3. Debug and test O/S changes.
4. Finish design of performance test cases.
5. Analyze results
6. Complete final report (writing needs time allocation as well)

Now... assign expected completion times and...

Proposal: Sample Schedule

ID	Task Name	Start	Finish	Duration	Sep 2005			Oct 2005			Nov 2005			Dec 2005			Jan 2006			Feb 2006			Mar 2006											
					18/9	25/9	2/10	9/10	16/10	23/10	30/10	6/11	13/11	20/11	27/11	4/12	11/12	18/12	25/12	1/1	8/1	15/1	22/1	29/1	5/2	12/2	19/2	26/2	5/3	12/3	19/3	26/3	2/4	
1	Requirements Elicitation	15/09/2005	01/10/2005	17d																														
2	Requirements Analysis	01/10/2005	01/11/2005	32d																														
3	System Design	01/11/2005	01/12/2005	31d																														
4	Object Design	15/11/2005	01/12/2005	17d																														
5	Implementation	01/01/2006	01/03/2006	60d																														
6	Testing	01/03/2006	24/03/2006	24d																														
7	Progress Report	21/11/2005	05/12/2005	15d																														
8	Changes to Oral Presentations Info Sheet	23/12/2005	06/01/2006	15d																														
9	Oral Presentation	09/01/2006	23/01/2006	15d																														
10	Changes to Poster Fair Info Sheet	25/01/2006	08/02/2006	15d																														
11	Poster Fair	27/02/2006	13/03/2006	15d																														
12	Draft of Final Report	10/03/2006	24/03/2006	15d																														
13	Final Project Report	24/03/2006	07/04/2006	15d																														

Proposal

Important (also for Final Report)

- For each team member, include a section/paragraph that describes how the project relates to your degree program

Fall Deliverable: Progress Report

- **Due: Dec 6**
- Purpose:
 - A mid-point checkup on the progress achieved, with reference to the original proposal.
 - A chance to re-define your project, now that you know more.
 - A (refined) plan for the final term.
 - It should be built on the proposal.
 - Copy and paste all of this into your final report.
- We will share the progress report with 2nd reader

Progress Report Structure

- Purpose: A mid-point **checkup on the progress achieved, with reference to the original proposal.**
- Other purpose:
 - Can also be used as initial step in oral presentation
 - Idea: Present a draft of your background and design
- Typical format: depends on amount of work done to-date and the development methodology you follow.
 - Waterfall Model: background and design chapters should be done (project is half-over)
 - Agile Development: some non-trivial product with a few key features

Progress Report Structure (cont.)

- Must make reference to your proposal
- Show clearly how much progress has been made
- Make a prediction as to how the rest of the project is likely to develop
 - Refine your schedule for the final term
- State any variation from the project proposal and the rationale that now seems necessary.
 - A chance to re-define your project, now that you know more

Oral Presentations in 2022/23

- **Time frame: Jan 20 – 24**
- Schedule more time per group member (20 instead of 15 minutes) to allow to incorporate demo
 - Live demo
 - Recording of project “at work”
- Scheduling will be done by project supervisor and the 2nd Reader
- Presentations can be done in different modes: in-person, online, or hybrid

Opportunities

- Competitions
 - IEEE Student Design Competition
 - <https://ieee-cas.org/cass-student-design-competition-0>
 - Innovative Designs for Accessibility (IDeA) Competition
 - <https://www.univcan.ca/programs-and-scholarships/innovative-designs-accessibility-competition/>
 - The Ontario Engineering Competition (See CSES)
 - The Canadian Engineering Competition (See CSES)
- Patents
- Departmental Toystore
 - We're looking for off-the-shelf demos that we can use in University Open Houses and other recruiting events
 - Will most likely have a video competition, details to be announced early in the Winter term
 - With cash prizes for winning team(s)

Opportunities (cont.)

- W.E. Cowie Innovation Award
 - Awarded by Faculty of Engineering and Design
 - Prize money \$27,000 in 2019/20
 - Winner in 2019/20: LEAP (a 4th year project)
 - Winner in 2018/19: Autonomous Vehicle (a 4th year project)
 - See also <https://carleton.ca/engineering-design/story/undergrad-autonomous-vehicle-project-wins-w-e-cowie-award/> for other competitions the group entered
 - Application Deadline: April 2025 or so, confirm with Dean's office

Cautionary Notes

- Plagiarism
 - Review your Professional Practice notes
 - If in any doubt, consult your supervisor before submitting any work (documents or code) that includes references.
- Health and Safety
 - This project will not cause RSI.
 - Projects involving human subjects require approval!

Next Meeting?

- TBD, may or may not be held, but expect it to be early-to-mid January if any
 - An email reminder will be sent
- Agenda
 - Oral Presentations
 - Poster Fair (if applicable)
 - Video and Final Report

Some Suggestions

- Start early
- Work smart
 - Dedicated time and self-management
 - Communications between/among members and with the supervisor(s)
 - Good time to learn and improve your communication skills
 - **ENGAGE:** active participation
 -



With **Freedom** comes ...
Responsibility.

Self-management is key:

- Time management
- Self motivation
- Stress management
 - Adaptability
- Decision making
- Goal alignment
- Personal development

Engage

*Every aspect,
starting from the beginning*

Protect yourself: **WEEDS**

- **W**: Wash hands often
- **E**: Eat healthy
- **E**: Exercise regularly
- **D**: Drink enough water/fluids
- **S**: Sleep well and keep you Soul/Spirit happy



When it seems impossible:

Give yourself a “space”

- Time
- Rest
- Exercise, walk
- Away from phone
- ...

Even by adding a tiny space:

Impossible becomes
Im possible
(I'm possible)

The Project is Yours to Explore

